MIL-H-3751B
19 March 1976
SUPERSEDING
MIL-H-3751A
11 December 1958

MILITARY SPECIFICATION

HARROW, DISK OFFSET, 8-FOOT,

24-INCH DISK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a tractor-drawn disk harrow.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-C-650	- Crates, Wood, Open and Covered.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-26195	- Boxes, Wood-Cleated, Skidded, Load-Bearing Base.

FSC 3710

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STANDARDS

FEDERAL

FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The disk harrow (hereinafter referred to as "harrow") shall consist of axle-mounted disks, an angling device, a leveling device, disk scrapers, weight boxes, and a tractor hitch.

3.2 First article (preproduction model). The contractor shall furnish one or more harrows for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce harrows that comply with this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4 Construction. The harrow shall be constructed as a tandem unit. The harrow shall withstand operation and service without malfunction or permanent deformation and shall meet the operational requirements specified

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herein. All components subject to wear, breakage, or distortion shall be accessible for adjustment and repair. The harrow shall conform to the following:

Cutting width - Not less than 8 feet.
Weight, ready for operation - Not less than 1800 pounds.

3.4.1 Disks. Each disk shall be 24 inches in diameter, and shall be of heat-treated steel not less than 0.165 inch thick. The disks shall be concave and shall be spaced to roll the earth from one furrow to the adjacent furrow. The disks shall be mounted on a steel axle which shall revolve on two or more antifriction bearings. The bearings shall be sealed against mud, dust, and water, and shall be of the permanently lubricated type.

3.4.2 Angling. An angling device shall be provided for changing the angle of the harrow in relation to the direction of travel. The angle shall be adjustable between 13 to 24 degrees. Provisions shall be made for automatically securing the angle of the disks in not less than 6 locked positions including both the parallel and maximum positions.

3.4.3 Leveling device. A means shall be provided to hold the harrow level and yet permit it to follow uneven ground.

3.4.4 Scrapers. The harrow shall be equipped with either adjustable or full blade steel disk scrapers, as specified (see 6.2).

3.4.5 Weight boxes. The harrow shall be equipped with one or more heavy steel weight boxes to hold and transport not less than 200 pounds of loose dirt.

3.5 Towing hitch. The harrow shall be equipped with a towing hitch which will permit backing and turning the harrow. The towing hitch shall accommodate a 2-inch diameter drawbar pin.

3.6 Harrowing. The harrow, towed by a tractor at 3 to 4 miles per hour (mph), shall plow not less than 8 inches deep in sandy loam soil.

3.7 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves, or of adjoining parts, except for adjustment, and without selection for fit or performance.

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3.8 Treatment and painting. The harrow shall be cleaned, treated, and painted in accordance with the contractor's standard practice.

3.9 Identification marking. Each harrow shall be permanently identified in accordance with MIL-STD-130. Marking shall be easily visible and shall be legible.

3.10 Workmanship.

3.10.1 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers and lockwashers shall be provided when necessary. All fasteners shall be correctly torqued and shall have full thread engagement.

3.10.2 Weld quality. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

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4.3 Preproduction inspection.

4.3.1 Examination. The preproduction harrow shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The preproduction harrow shall be tested as specified in 4.5.2.1. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for the minor defect.

4.4.3 Test. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The harrows shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Cutting width not as specified.
103. Diameter of disks not as specified.
104. Angling device not as specified.
105. Leveling device not as specified.
106. Scraper not as specified.
107. Weight boxes not as specified.
108. Towing hitch not as specified.
109. Parts not interchangeable as specified.
110. Workmanship not as specified.

Minor

201. Identification marking incomplete, illegible, or not as specified.

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4.5.2 Test.

4.5.2.1 Operational. The harrow shall be subjected to an operational test of not less than 8 hours of operation at a speed of 3 to 4 mph in sandy loam soil not less than 8 inches deep. The test shall include operation of the harrow with disks angled in all positions. The harrow shall be operated for not less than 5 minutes at each angle. Concurrent with this test, the effectiveness of the leveling device, scrapers, and hitch shall be determined. Nonconformance to 3.4 through 3.6 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Tests. The preproduction pack of the crated or boxed harrow for Level A shall be subjected to one of the following tests. The railroad car impact test shall be used when such facilities are available. The pendulum impact test is an alternate test. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visible damage to the contents shall constitute failure of the test and shall be cause for rejection.

4.6.1.2.1 Railroad car impact. Subject the crated or boxed harrow to the railroad car impact test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars, with draft gear extended and the brakes set, at a speed of not less than 10 nor more than 11 miles per hour.

4.6.1.2.2 Pendulum impact. Subject the crated or boxed harrow to the pendulum impact test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following major defects. AQL shall be 4.0 percent defective.

111.	Materials, methods, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
112.	Disassembly not as specified for Level A.
113.	Maintenance tools not preserved and packaged in accordance with referenced documents for Level A.
114.	Consolidation not as specified for Level A.
115.	Strapping not as specified for Level A.
116.	Marking illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and testing within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be prepared utilizing either the preproduction model or a production harrow. When the preproduction model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the preproduction model and production harrows.

5.2 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss and to accomplish a reduction in cube. Removed bolts, nuts, pins, screws, and washers shall be reinstalled in one of the mating parts and secured to prevent their loss.

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5.2.1.2 Matchmarking. When necessary to facilitate reassembly, parts removed and mating parts shall be matchmarked. Matchmarking shall be printed on tags in accordance with UU-T-81, Type A. Printing thereon shall be with waterproof ink.

5.2.1.3 Unprotected surfaces. Unprotected metal surfaces requiring the application of a preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative in accordance with MIL-P-116.

5.2.1.4 Technical publications. When technical publications are required (see 6.5), they shall be preserved in accordance with MIL-P-116, Method IC-1.

5.2.1.5 Repair parts. Such repair parts as are provided (see 6.6) shall be preserved in accordance with the applicable provisions of MIL-P-116.

5.2.1.6 Maintenance tools. The tools provided (see 6.6) shall be preserved in accordance with the Level A requirements of PPP-P-40.

5.2.1.7 Consolidation. The disassembled components, technical publications, repair parts, tools, and any loose small components shall be consolidated in a box conforming to PPP-B-601, Overseas Type, Grade E, or PPP-B-621, Class 2. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Commercial. The harrow shall be preserved in accordance with FED. STD. No. 356.

5.3 Packing. Packing shall be Level A or Commercial, as specified (see 6.2).

5.3.1 Level A. Each complete harrow shall be packed in an open crate conforming to PPP-C-650, Type V, Style A, Class 1, or a box conforming to MIL-B-26195, Type II, Class 2, style optional, with rubbing strips. Strapping for closure and anchoring of contents shall be as specified in 5.2.1.7.

5.3.2 Commercial. Each complete harrow shall be packed in accordance with FED. STD. No. 356.

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5.4 Marking.

5.4.1 Level A. Level A marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Commercial marking shall be in accordance with FED. STD. No. 356.

6. NOTES

6.1 Intended use. The harrow is intended for use in the mixing of aggregate and bituminous materials with subgrade soils in progressive stabilization of military roadbeds.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model, and number of harrows required (see 3.2).
(c) Whether full blade or adjustable scrapers are required (see 3.4.4).
(d) Time frame required for submission of preproduction pack (see 5.1).
(e) Degree of preservation and packing required (see 5.2 and 5.3).
(f) When strapping shall be other than as specified (see 5.2.1.7 and 5.3.1).

6.3 Preproduction model. Any changes or deviations of production harrows from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish harrows conforming to this specification.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the harrows in accordance with this specification.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each harrow.

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6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of harrows.

Custodians:
Army - ME
Navy - YD
Air Force - 84

User activity:
DSA - CS

Preparing activity:
Army - ME

Project No. 3710-0006

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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